



M004 Week 7 — English Worksheet

Topic: Auto-Building Spaces — Loops + Functions + Chat Commands · **Course:** Functions & Games ·

Time: about 45 minutes

This week one chat command builds a **whole space by itself**. A loop repeats a room piece, a function builds each piece, and the **number argument** decides how big the space grows.

1 · Predict 🧠

Read each set of steps. Before you imagine the build happening, write what you think will happen.

```
on chat command "dungeon" with number rooms:
  repeat rooms times:
    build_room()
    move forward 6
```

You type **dungeon 3** in chat. How many rooms get built? What does **move forward 6** do between rooms?


```
function build_room():
  build_wall(5)
  build_wall(5)
  build_wall(5)
  build_wall(5)

on chat command "dungeon" with number rooms:
  repeat rooms times:
    build_room()
    move forward 6
```

`build_room` reuses `build_wall` four times. You type `dungeon 2` — how many walls get built in total?


```
on chat command "towers" with number count:
  repeat count times:
    build_tower(random number from 3 to 8)
    move forward 4
```

You type `towers 5`. Are all five towers the same height? Why does the row of towers feel random?

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — `dungeon 4` should build four rooms in a row. Right now all four rooms get built **in the same spot**, so it looks like only one room.

```
on chat command "dungeon" with number rooms:
  repeat rooms times:
    build_room()
```

Hint: after each room, the builder needs to move before building the next one.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The number you type should decide how many rooms get built. But `dungeon 5` and `dungeon 1` both build just **one** room.

```
on chat command "dungeon" with number rooms:
  repeat 1 times:
    build_room()
    move forward 6
```

Hint: the argument `rooms` is never used.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — Each room is **6 blocks long**. The rooms should sit side by side, but every new room breaks the back wall of the room before it.

```
on chat command "dungeon" with number rooms:
  repeat rooms times:
    build_room()
    move forward 5
```

Hint: compare the room length with the move distance.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now switch to your homework world. Make a chat command with a number argument that auto-builds a space — a loop should repeat a room piece. When you finish, come back here.

Send a photo or video of your space being built, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

My chat command is called ...

The number argument decides ...

The loop repeats ...

Between each piece, the builder ...

One tricky moment was when ...

If I had more time, I would ...

4 · Record Your Walkthrough

Now take a video on your phone (or a parent's phone) while your command builds the space. Talk through it like you are teaching someone who has never seen it. Try to use these words: **chat command, argument, loop, repeat, function**.

1. Show the empty space, then type your command with a **small** number.
2. Say what the number argument changed.
3. Run the command again with a **bigger** number and show the difference.
4. Say in your own words how the loop and the function work together.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.